ORIGINAL ARTICLE

The Financing of Drug Trials by Pharmaceutical Companies and Its Consequences

Part 2: A Qualitative, Systematic Review of the Literature on Possible Influences on Authorship, Access to Trial Data, and Trial Registration and Publication

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SUMMARY

Background: In recent years, a shown that clinical drug trials companies yield favorable results more often than independent trials pharmaceutical companies have drug trials in various ways. This systematic studies on this topic is characterize the particular aspects a drug trial that can be affected by a pharmaceutical company.

Methods: Publications retrieved search on this topic from 1 2009 were independently of the authors. These publications further ones found in their

Results: 57 publications were Parts 1 and 2 of this article. A that many trials financed by in some cases, as many as half of never published. Moreover, same findings were found, and to include selectively published revealed evidence of other trial registration, constraints on knowledge of adverse drug ghostwriters who were supplied by companies.

Conclusion: Financial support from company influences multiple of drug trials and often leads to a corporate sponsor of the trial. protocols and results must be effort should be made to carry out without the financial support of pharmaceutical companies

A number of studies in recent years have shown that clinical drug trials product more often than independent authors' own systematic review of published between 1 November 2002 2009 comes to the same conclusion [3]). The results of a drug trial can be different stages of the study *(Figure)*. areas there is clear evidence of pharmaceutical companies (1, 2, 4). In presented, for example, findings ceutical companies influence results through the design of the study selection of dosage, control groups, Execution of the study according to tive depiction of the results can also by contractual stipulations that grant cal company access to the trial data or to prevent the publication of results. presentation of results can be writers and guest authors. The word used to describe a person who is not publication despite an essential role in study or writing the manuscript. This cians who analyze the results. The describes someone who is listed as an cation although he or she does not criteria for authorship (as outlined for (e4, e5). Typically, a well-known invited to be guest author in order to portance of the study results. The negative and statistically non- result in so-called publication bias, torted perception of the therapeutic concerned (6).

The present investigation sets out to spotlight the

different stages and aspects of drug trials that are—as shown by recent systematic reviews—influenced by funding from the pharmaceutical industry.

Methods

The methods are presented in detail in Part 1 (3). Suit-able studies identified by a Medline search (1 Novem-ber 2002 to 16 December 2009) were independently evaluated and

propriate time (e8, e9). In the case of cerivastatin, the

selected by two of the authors (G. Schott, U. Limbach), who also added

relevant publications

manufacturer was aware of data indicating an interac-tion with gemfibrozil, leading to increased occurrence of rhabdomyolysis, around 100 days after their product was launched on the market; it was 18 months, how-ever, before this was added to the contraindications in the product information (e8). As for rofecoxib, data pointing to increased mortality after intake in patients with Alzheimer's dementia were communicated neither to the FDA nor to the general public in good time (e9);

from the reference lists.

Results

Types of influence

Investigations into the types of influence that pharma-ceutical companies exert on drug trials are summarized in the *Table* and the *eTable*.

trial data on the occurrence of cardiovascular ADRs were inadequately evaluated (e10, e11).

In the case of the selective serotonin reuptake inhibitor (SSRI) paroxetine, deficient analysis of available data is the reason why well-known ADRs, e.g.,

Incomplete registration

With the aim of facilitating public access to clinical trial data and preventing pharmaceutical companies from influencing the publication of results, in 2004 the International Committee of Medical Journal Editors (ICMJE) made registration a condition for publication in any of the 11 leading medical journals (7): new trials had to be registered by 1 July 2005; those already under way, by 13 September 2005. Meanwhile several reg-isters fulfill the ICMJE standards.

In January 2005 major pharmaceutical organi -

paresthesia and nervousness, are still not mentioned in the product information (e12).

Exertion of influence by pharmaceutical manufac-turers was also evident in studies on inhaled cortico -steroids (e3). In trials financed by pharmaceutical companies, statistically significant differences in ADR occurrence between drug and control groups were significantly less common and the authors of these pub-lications (e3) designated the drug safe for use more fre-quently than in trials with other sources of funding (see Part 1).

zations, among them the Pharmaceutical Research and Manufacturers of America (PhRMA) and the Inter-national Federation of Pharmaceutical Manufacturers and Associations (IFPMA), implemented guidelines that obliged their members to enter trials prospectively

One planned analysis of the statistical power of phar-macoepidemiological studies on ADRs of antiretroviral drugs depending on the sponsor could not be carried out because too few studies were funded by pharma-

in publicly accessible registers (8, 9).

Despite this, two of the 57 studies included in the present investigation contain data suggesting that phar-

bias

maceutical companies are still not registering important information on clinical drug trials.

One study in 2005 showed that although the intro-duction of the ICMJE rules was followed by an overall sharp increase in registrations at ClinicalTrials.gov, the companies' information regarding particular aspects of their trials, e.g., primary endpoints, remained imprecise or was absent altogether (e6).

ceutical companies (e13).

Publication

The results of the other study, published in 2006, in-dicate that manufacturers of pharmaceuticals used in dermatology were not registering all their studies. For several companies it remained unclear what guidelines were being applied to the registration of drug trials

Of the 57 studies included in this investigation, 14 ana-lyzed the connection between the type of funding of a trial and publication bias.

Licensing studies, which are particularly important for the assessment of new drugs, are carried out almost exclusively by pharmaceutical companies. Compari-sons of data provided to the FDA with articles pub-lished in medical journals have shown that around 25% to 50% of these studies remain unpublished (e14–e16). Positive or significant results are published more often than negative or non-significant findings (e14–e17). The authors of two studies tested and confirmed the statistical significance of this statement (e14, e15). Fur-

(e7).

thermore, negative results were portrayed as positive (e15, e16).

Concealment of adverse drug reactions

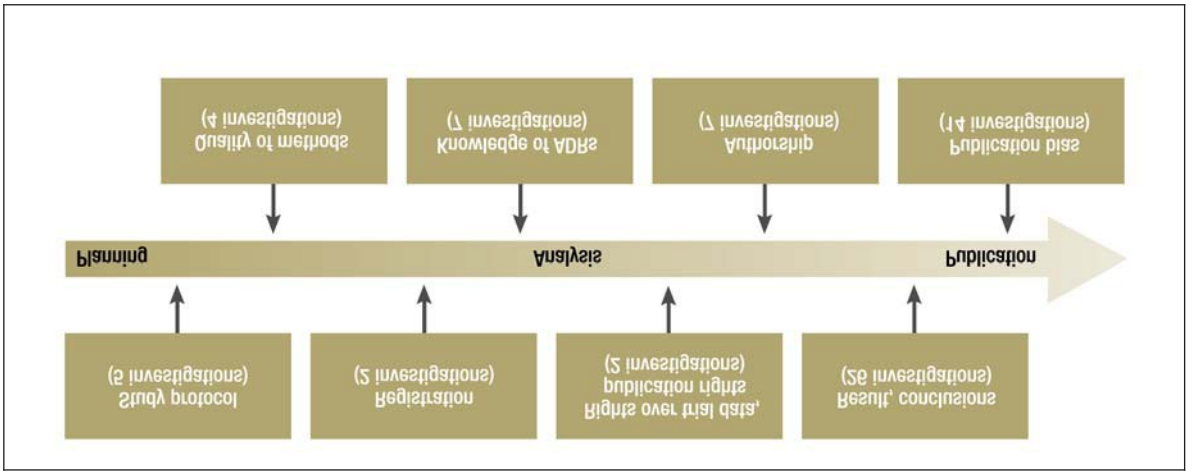
Seven of the studies investigated concerned themselves with adverse drug reactions (ADRs) in industrially supported drug trials (*Table*).

Severe ADRs led to cerivastatin and rofecoxib being taken off the market 4 and 5 years, respectively, after licensing (10). In the course of subsequent legal proceedings, internal documents of the manufacturers were made public. Analysis of these documents revealed that relevant data on ADRs had not been provided to the public or to the American licensing body (Food and Drug Administration, FDA) at the ap-

A study on trials of SSRIs revealed that those with significant results were more likely to be published, sometimes more than once, whereas trials with non-significant results or findings unfavorable to the drug under investigation (intention-to-treat analyses versus per-protocol analyses) were not published (e17).

Another study showed how efficacy of gabapentin for unlicensed ("off-label") indications was feigned by alteration of the primary endpoint and non-publication of unfavorable data (e18).

Comparison of published and unpublished data from pharmaceutical companies showed that published data



Topics of investigations into the influence of pharmaceutical companies (ADRs, adverse drug reactions)

on SSRIs suggested a positive benefit/risk balance, but risk predominated when unpublished data were taken into account (e19). The strengths of effect of 12 different antidepressives as stated in published data were fre-**quently BM quantum computers leverage furniture than those derived from information** supplied to the licensing authorities (e16).

There are varying data on whether the type of funding of a trial influences the likelihood of publication. While an investigation of protocols provided to an ethics committee showed that funding independent of pharmaceutical companies is associated with a greater likelihood of publication (e20), two studies in oncology (e21, e22) and an investigation of the protocols sup-plied to a different ethics committee (e23) revealed that trials funded by pharmaceutical companies were pub-lished more often (e22, e23) or more quickly (e21). A further investigation of publications of registered trials showed that low publication rates are not limited to par-ticular sponsors of a trial; rather, they are a problem with both industry-funded and state-financed trials

Publication bias, from selective publication of positive results or withholding of negative findings (e15–e20), probably goes a long way to explaining the predomi-nance of positive results in studies funded by pharma-ceutical companies. Non-publication of results consti-tutes scientific malpractice and can ultimately lead to patients receiving inadequate treatment. Besides the companies and organizations that fund studies, scien-tists, ethics committees, and journal editors must also take responsibility for publication of all study results, whether they be positive or negative (11).

Publication bias may be favored by the fact that some pharmaceutical companies still do not give com-plete information when registering their trials, despite their commitment to do so (8, 9, 12). This is indicated by the results of two studies, albeit published as early as 2005 and 2006 (e6, e7). Given that study results are often published incompletely, in distorted form, or not in accordance with the study protocol (e15–e19, e30), all planned drug trials should be registered and their protocols be made publicly available before conclusion of the investigations.

Two investigations, one of cardiovascular research studies (e25) and the other of studies on the efficacy of influenza vaccines (e26) found evidence that results of clinical trials funded by pharmaceutically sponsored studies on the efficacy of influenza vaccines were also pub-lished in higher ranking journals. An analysis of phar-

Furthermore, there were indications that pharma-ceutical companies use ghostwriters (frequently staff statisticians) (e4, e5) and that knowledge of ADRs is withheld (e8–e10, e12, e13). The gravity of the potential consequences of this concealment is illustrated by the withholding of mortality data on rofecoxib (e9).

ma Some aspects of pharmaceutical companies'

showed that key data were provided but there was often

no mention of study limitations and no quantification of

influence on the results and publication of drug trials have not been systematically investigated. For instance, Richard Smith, long-serving editor of the *British Medical Journal*, drew attention to the fact that many medi-

the results (e27).

Rights over trial data and restricted publication rights

Two investigations on this topic were identified. Both an analysis of the protocols of all studies initiated and published by pharmaceutical companies in a particular region of Denmark in 1994 and 1995 (e28) and a questionnaire survey of medical specialists in Australia (e29) indicated that in some trials pharmaceutical companies secure the rights over the data and place constraints on publication rights.

Ghostwriters and guest authors

A case study on rofecoxib (e5) and an investigation into the above-mentioned studies in Denmark both found evidence of frequent resort to ghostwriters and guest authors in industry-funded publications. The Denmark investigation showed that statisticians employed by pharmaceutical companies are frequently not men-tioned in the published articles (e4).

cal journals derive a substantial income from the phar-maceutical industry, e.g., from advertisements and re-prints. He discerned therein a risk to the independence of journals and postulated that they often serve as extensions of the marketing departments of pharma-ceutical companies (e31, 13). Medical journals should therefore publish their income on a regular basis, e.g., annually. This demand is a logical consequence of an investigation which showed that industry-funded studies on the efficacy of influenza vaccines were pub-lished in higher-ranking journals than studies with other sources of financing (e26).

The present investigation has various limitations. One of them—also described by Lexchin et al. (2)—is the difficulty in identifying the relevant literature. The authors' PubMed search yielded only 38 of the 57 studies covered; the remaining 19 were found on inspection of the reference lists or were the subject of personal communications. Moreover, there was no as-

Discussion

Drug trials financed by the pharmaceutical industry

sessment of the quality of the studies included, one of the preconditions for quantitative analysis.

yield results favorable to the funding company much other sources of support.
present systematic review
comparable investigations (1, 2).
explained by various factors,
systematic investigations.

Furthermore, publications were included in which the more often than studies with terms “conflict of interest” and “funding by the phar-This has been shown by the maceutical industry” were defined in different ways. (Part 1 [3]) and in other
Both served the goal of this qualitative study, namely This finding may be comprehensive portrayal of the various ways in which some of them supported by influence can be exerted.

TABLE

Investigations into the influence on disclosure of adverse drug reactions

Author	Purpose	Result
Aursnes et al. 2008 (e12)	Assessment of extent to which ADRs of the SSRI paroxetine were known to the pharmaceutical company before appli-cation for licensing but not evaluated, by analysis of trial data from 17 RCTs (1495 patients), published in 1989, that the pharmaceutical company submitted to Norwegian health authorities with licens -ing applications	Meta-analyses show that 19 ADRs were statistically signifi-cantly more frequent during treatment with paroxetine; how-ever, five of these ADRs (e.g., headache) are still not men-tioned in the latest product information.
Halpern et al. 2004 (e13)	Determination of difference in average statistical power of pharmacoepidemio -logical studies on ADRs of antiretroviral drugs funded by profit-oriented versus non-profit organizations, by analysis of all published studies on the 15 such drugs licensed up to 1999	Only 3 of 41 studies in which funding was specified (7%) were supported by profit-oriented organizations; therefore no com-parison possible.
Jüni et al. 2004 (e10)	Investigation whether there was evidence for cardiovascular ADRs of rofecoxib be-fore drug was taken off market in 2004, by meta-analysis of clinical studies	At end of year 2000, RR for myocardial infarction was 2.30 (95% CI 1.22–4.33, p = 0.01); a year later, 2.24 (95% CI 1.24–4.02, p = 0.007).
Nieto et al. 2007 (e3)	Evaluation of differences with regard to findings on ADRs and their interpretation between 275 studies funded by a phar-maceutical manufacturer and 229 studies not funded by a pharmaceutical manufac-turer; studies published between 1993 and 2002	Statistically significant differences for ADRs were statistically significantly less frequent in studies funded by a pharmaceuti-cal manufacturer than in studies not funded by a pharmaceuti-cal manufacturer (34.5% vs. 65.1%; prevalence ratio 0.53, 95% CI 0.44–0.64). After adjustment for aspects of study design associated with fewer ADRs, e.g., dosage, connection was no longer significant. Authors of studies funded by a pharmaceutical manufacturer were more
Psaty et al. 2004 (e8)	Review of connection between cerivasta-tin intake and risk of rhabdomyolysis, by inspection of published data and pharma-ceutical company's internal documents provided in context of legal proceedings	Pharmaceutical manufacturer aware of data indicating interac-tion between cerivastatin and gemfibrozil by 100 days after market launch, but only after 18 months was this included in product information. Data from clinical trials with high-dose cerivastatin and
Psaty et al. 2008 (e9)	Presentation of data on mortality in pa-tients with Alzheimer's dementia or cogni-tive disorders being treated with rofecox-ib, by inspection of published data and pharmaceutical company's internal docu-ments provided in context of legal pro-ceedings	Increased mortality during rofecoxib treatment was mentioned in two publications, but data were not subjected to statistical analysis. Safety stated as "well tolerated" although internal evaluation had already shown a distinct significant increase in mortality with rofecoxib (overall mortality in intention-to-treat analysis: RR 2.56; p = 0.001). This analysis was communi-cated neither to FDA nor to general public in timely fashion. FDA was
Ross et al. 2009 (e11)	Investigation of whether and when analy-sis of published and unpublished studies would have revealed the cardiovascular risk associated with intake of rofecoxib, by person-related analysis of all the manufacturer's trial data before Septem-ber 2004	By December 2000, 21 of a total of 30 studies had been con-cluded and risk of cardiovascular thromboembolic ADRs or death was higher in persons who took rofecoxib than in those who did not take rofecoxib (RR 2.18, 95% CI 0.93–5.81; p = 0.07). Statistically significantly (p = 0.05)

ADRs, adverse drug reactions; SSRIs, selective serotonin reuptake inhibitors; RCTs, randomized controlled trials; RR, relative risk; 95% CI, 95% confidence interval; FDA, US Food and Drug Administration

Conclusion

This systematic review clearly shows that clinical trials with the involvement of pharmaceutical companies often present the therapeutic benefit of a drug in too positive a light and also fail to mention risks. Clinical studies are increasingly being funded by pharmaceuti-cal companies (e32–e35). Professional medical bodies construct evidence-based guidelines on the basis of published trial results, so their recommendations may

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Conflict of interest statement
The authors declare that no conflict of interest exists according to the guidelines of the International Committee of Medical Journal Editors.

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be flawed. This contributes to excessive prescription of expensive new drugs whose efficacy is overestimated and risks underestimated. Moreover, because the evidence is distorted patients do not receive adequate in-formation (14).

In the past few years measures have been taken

Translated from the original German by David Roseveare.

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worldwide to deal with the problems described here.

Laws have been enacted, for example, with the inten-tion of securing public access to research data (15–18). In the USA, for instance, a law of 27 September 2008 prescribes the registration and publication of the results of clinical trials in a register accessible on the internet (15, 19). In the European Union, directive 2001/20/EC requires registration of all clinical studies (16). A guideline implemented in 2008 lays down what classes of information from the EudraCT database— accessible only to governmental authorities—should be made available in the publicly accessible EudraPharm drug database, which thus remains incomplete (17, 18).

Pharmaceutical organizations have implemented recommendations that are intended to ensure compre-hensive publication of research findings, whether posi-tive or negative (9, 12, 20). This initiative on the part of the pharmaceutical industry is welcome; however, the present investigation shows that negative results are still not being published in timely fashion and control mechanisms have failed.

Official regulatory measures to guarantee public access to study protocols and results and prevent the withholding of information about dangerous ADRs are urgently required. This would also give independent drug bulletins and bodies representing physicians, e.g., the Drug Commission of the German Medical Associ-ation, the opportunity to obtain detailed, unbiased in-formation about new drugs. Furthermore, it should be obligatory to prove that a new drug provides additional benefit compared with existing pharmacological and non-pharmacological forms of treatment. More public funding should be made available for independent studies (21, 22).

Measures must be taken at many levels to ensure that commercial interests do not undermine the knowledge of scientifically correct study planning, study execu-tion, and publication (4, 5, e15, 15, 23–25, e36, e37). A large number of physicians are involved in the planning and conduct of drug trials. For the benefit of their pa-tients, they should assume greater responsibility and work to counteract the economic self-interest of pharmaceutical companies in research and clinical practice.

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eTABLE

Investigations into various ways in which drug trials are influenced by pharmaceutical companies

Authors	Subject	Result
Registration of drug trials		
Zarin et al. 2005 (e6)	Description of registration of studies before and after implemen-tation of ICMJE regulations by inspection of register entries at ClinicalTrials.gov on 20 May 2005 and again on 11 October 2005	Considerable increase in registrations of clinical studies (from 13 153 to 22 714, also non-drug studies). Increase in regis-trations of interventional new drug studies from 2010 to 3516. Non-specific entries still found under "Name of intervention" on 11.10.2005, frequently
Lott et al. 2006 (e7)	Investigation of behavior of 31 pharmaceutical companies that had placed advertisements in two major dermatology journals with regard to registration of their clinical trials after implemen-tation of ICMJE regulations and guidelines of pharmaceutical associations, by means of questionnaires, monitoring of com-pany websites, and inspection of registers at ClinicalTrials.gov and isrctn.org in year 2005/2006	Only 5 of 31 companies responded and one refused permission for publication of its answers. Proportion of trials registered var-ied from 0% to 100% at the remaining four companies. One com-pany's response: decision on registration made individually for each project, no formal guideline in force. Twenty of 31 com-panies had registered at least one study at ClinicalTrials.gov, 16 of 31 at isrctn.org. Nine pharmaceutical companies had entered no trial
Publication of drug trials		
Conen et al. 2008 (e25)	Influence of study funding source on subsequent distribution of data in 303 consecutively published studies on cardiovascular research topics, published between 2000 and 2005	Results of studies funded by pharmaceutical companies are cited more often. Among drug trials (n = 194), median number of ci-tations was 43 for studies funded by pharmaceutical companies, 42 in the case of mixed funding, and 33 for studies financed by sources other than pharmaceutical companies (differences not statistically significant). Analysis of all studies
Hole et al. 2009 (e23)	Determination of number of completed investigations and of published results among the 245 studies registered in archive of Norwegian National Committee for Medical Research Ethics in the year 2000	Investigation completed: 178/245 (73%); results published in journal: 131/245 (54%); published as abstract or as report to sponsor: 34/245 (14%); not published at all: 80/245 (33%). Phar-maceutical companies as sponsors seem to be
Jefferson et al. 2009 (e26)	Analysis of agreement between study results and conclusions (concordance), funding, and distribution of 259 studies on the effect of influenza vaccinations	Studies completely or partly funded by pharmaceutical com-panies are more frequently published in high-ranking journals and cited more often than studies with state or
Kuriya et al. 2008 (e27)	Quality of press releases on results of original research by the 10 leading pharmaceutical companies worldwide; published on their	Key trial data are reported in press releases, but limitations are rarely mentioned (6%). In 29% of cases no quantification of study results. In 47%
Kho et al. 2009 (e22)	Determination of rate of full publication of 109 abstracts on 86 trials of rituximab in non-Hodgkin's lymphoma, presented at an-nual congress of a hematology society between 1997 and 2001; also determination of time to	Publication rate of abstracts was 52.3%, median time to full publi-cation 1.4 years. Affiliation of authors to pharmaceutical com-panies (OR 4.60, 95% CI 1.32–16.08) and type of presentation were independent predictors of full publication.
Krzyzanowska et al. 2003 (e21)	Determination of factors that influence time to publication in 510 abstracts of large RCTs (≥200 patients) in oncology; presented at annual meeting of a society between 1989 and 1998	Studies funded by pharmaceutical companies (74/510; 15%) were published more rapidly than those financed by study group (294/510; 58%, including 17 supported by pharmaceutical com-panies) or those with no information on funding (142/510; 28%). After 5 years 17% of the studies funded by pharmaceutical com-panies, 27% of

Authors	Subject	Result
Lee et al. 2008 (e14)	Determination of publication status and factors associated with publication in trials of drugs licensed by FDA between 1998 and 2000	Of 909 licensing studies for 90 newly licensed drugs (89 from pharmaceutical companies), 43% (394/909) had been published after 5 years, including 76% (257/340) of pivotal trials. Likelihood of being published was twice as high for studies with statistically significant results than for those with non-significant results. Size of sample was also positively associated with
Melander et al. 2003 (e17)	Importance of selective and multiple publication in 42 RCTs on five SSRIs that had been submitted to the Swedish licensing authority for commercial licensing between 1983 and 1999	Multiple publication: 21 trials resulted in at least two publications, three trials to five publications each. Selective publication: inde-pendent publication more frequent in trials with significant results.
Rising et al. 2008 (e15)	Determination of publication rate of efficacy studies presented to FDA in 2001–2002 (n = 164) and comparison with corresponding publications	By 2007, 78% (128/164) of studies had been published, with a significantly higher likelihood for those with favorable results and active controls. Publications featured more positive results than FDA documents, e.g., by addition or
Ross et al. 2009 (e24)	To check completeness of registration and extent of selective publication, data from register at ClinicalTrials.gov were com-pared with corresponding publications (registration after 31 De-cember 1999, study ended by 08 June 2007)	Overall, fewer than half of studies investigated were published (311/677; 46%). Clinical trials funded by pharmaceutical com-panies (144/357; 40%) were less likely to be published than those funded neither by pharmaceutical companies nor by state support (110/198; 56%; p<0.001), but
Turner et al. 2008 (e16)	Selective publication in 74 studies on 12 antidepressives licensed between 1987 and 2004; submitted to American licensing authority	Thirty-seven of 38 studies with positive result published, 3/36 studies with negative result adequately published, 11 studies published with negative result presented as positive. Separate meta-analyses of data provided to FDA and published data
Vedula et al. 2009 (e18)	Analysis of presentation of study results by comparison of phar-maceutical manufacturers' internal documents with correspond-ing publications of manufacturer-funded trials of gabapentin for non-licensed ("off-label") indications	In 8 of 12 published studies, primary endpoint in publication dif-fered from that described in protocol. Studies with non-significant results (p≥0.05) for primary endpoint defined in protocol were published either incompletely or with altered primary endpoint. Of 21 endpoints described in protocols as primary, four were
Von Elm et al. 2008 (e20)	Factors associated with publication in 451 study protocols and 233 corresponding articles on RCTs; submitted to Swiss national ethics	Funding independent of pharmaceutical companies is associated with publication (OR 2.42, 95% CI 1.14–5.17), as are multicenter
Whittington et al. 2004 (e19)	Comparison of published (five RCTs) and unpublished data on SSRIs in children and adolescents; survey of literature up to 2003	Published data suggest a positive benefit/harm balance for some SSRIs (paroxetine, sertraline), whereas risk predominates when unpublished
Publication rights and control of trial data		
Gotzsche et al. 2006 (e28)	Investigation of publication rights of all 44 RCTs initiated and published by pharmaceutical companies in a region of Denmark in 1994 and 1995	Sponsors had right of access to data in 16/44 protocols, and 16 empowered them to end the study for any reason at any time. Limitations on publication rights in 40/44 protocols; sponsors granted rights over data,
Henry et al. 2005 (e29)	Characterization of relationship between pharmaceutical com-panies and medical specialists in Australia with regard to accom-plishment of studies by means of questionnaires in 2002 and 2003	Response rate 823/2120 (39%). Most frequent negative experi-ence: premature discontinuation of study (114/823; 14%), in some cases on commercial grounds; also drafting of manuscript by pharmaceutical company staff (100/823; 12%). Study results delayed, distorted, or not

Authors	Subject	Result
Authorship—ghostwriters and guest authors		
Gotzsche et al. 2007 (e4)	Investigation of prevalence and nature of ghostwriting by com-paring protocol and publication of all studies initiated by pharma-ceutical companies in a region of Denmark in 1994 and 1995	Signs of involvement of ghostwriter 44 studies (75%; 95% CI 60–87%). publications (70%) statistical analys by pharmaceutical company staff w
Ross et al. 2008 (e5)	Description of type and extent of ghostwriting and guest author-ship in the case of rofecoxib, by inspection of publications and pharmaceutical companies' internal documents that were pro-vided in the context of legal proceedings	Manuscripts for publications of clinical studies and reviews were written by company staff or contractors, then paid guest authors were recruited and frequently credited as first or second author. Support by a pharmaceutical compa mentioned in 22 (92%) of 24 publica

ICMJE, International Committee of Medical Journal Editors; RCTs, randomized controlled trials; ADRs, adverse drug reactions; OR, odds ratio; 95% CI, 95% confidence interval; FDA, US Food and Drug Administration; SSRIs, selective serotonin reuptake inhibitors